Matthew Tran
Configuring GRUB Security

I secured the GRUB bootloader on Ubuntu by hiding its menu, creating a privileged superuser and a restricted regular user, then verified and applied the changes through a reboot. This hardened the system against unauthorized startup access.

Tools I used…

- Ubuntu Virtual Machine
- GRUB Bootloader
- gedit

Some skills I learned are …

- Configuring GRUB settings to improve system security
- Hiding the GRUB menu to prevent unauthorized boot access
- Creating a GRUB superuser with full administrative privileges
- Creating a regular user with limited access
- Using Linux commands to manage users and permissions
- Verifying security changes through system files and reboot testing

export PS1='matthew_tran@vm:\w$ '

Personalizes the terminal

Task 1) Hiding the GRUB Menu and Setting Automatic Boot

```
matthew_tran@vm:~$ sudo -H gedit /etc/default/grub

(gedit:3056): Tepl-WARNING **: 20:45:20.859: GVfs metadata is not s
upported. Fallback to TeplMetadataManager. Either GVfs is not corre
ctly installed or GVfs metadata are not supported on this platform.
 In the latter case, you should configure Tepl with --disable-gvfs-
metadata.
matthew_tran@vm:~$
```

sudo -H gedit /etc/default/grub
This opens the GRUB configuration for editing. This file controls GRUB's behavior during boot; changes here will secure the system.

```
# If you change this file, run 'update-grub' afterwards to update
# /boot/grub/grub.cfg.
# For full documentation of the options in this file, see:
#   info -f grub -n 'Simple configuration'

GRUB_DEFAULT=0
GRUB_TIMEOUT_STYLE=hidden
GRUB_TIMEOUT=0
GRUB_DISTRIBUTOR=`lsb_release -i -s 2> /dev/null || echo Debian`
GRUB_CMDLINE_LINUX_DEFAULT="quiet splash"
GRUB_CMDLINE_LINUX=""

# Uncomment to enable BadRAM filtering, modify to suit your needs
# This works with Linux (no patch required) and with any kernel that obtains
# the memory map information from GRUB (GNU Mach, kernel of FreeBSD ...)
#GRUB_BADRAM="0x01234567,0xfefefefe,0x89abcdef,0xefefefef"

# Uncomment to disable graphical terminal (grub-pc only)
#GRUB_TERMINAL=console

# The resolution used on graphical terminal
# note that you can use only modes which your graphic card supports via VBE
# you can see them in real GRUB with the command `vbeinfo'
#GRUB_GFXMODE=640x480

# Uncomment if you don't want GRUB to pass "root=UUID=xxx" parameter to Linux
#GRUB_DISABLE_LINUX_UUID=true

# Uncomment to disable generation of recovery mode menu entries
#GRUB_DISABLE_RECOVERY="true"

# Uncomment to get a beep at grub start
#GRUB_INIT_TUNE="480 440 1"
```

GRUB_DEFAULT=0

This line makes sure that the boot option is default to the first operating system within the GRUB menu. It makes sure that the default operating system is automatically chosen when the system boots up, which will lead to system stability and security.

GRUB_TIMEOUT_STYLE=hidden

This will hide the GRUB menu when the boot starts. This will reduce the risk of any unauthorized users messing with the boot process.

GRUB_TIMEOUT=0

This sets the timeout to 0 seconds, which means that the system will boot without displaying anything. This is important and useful because not only will it be safer from people who might want to tamper with the boot menu, but it also saves a lot of time considering that the boot menu will most likely not be used every time you need to start your system.

Task 2) Creating a Superuser Account for GRUB

```
matthew_tran@vm:~$ sudo adduser super
Adding user `super' ...
Adding new group `super' (1001) ...
Adding new user `super' (1001) with group `super' ...
Creating home directory `/home/super' ...
Copying files from `/etc/skel' ...
New password:
```

sudo adduser super

This step adds a new user named “super.” The user will be given many abilities to configure with the bootloader settings.

```
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for super
Enter the new value, or press ENTER for the default
        Full Name []:
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n]
matthew_tran@vm:~$
```

PWD = noaccess

This assigns a simple password. Although simple, it is another layer of protection.

```
matthew_tran@vm:~$ sudo usermod –aG sudo super
```

sudo usermod –aG sudo super

This command will add the user “super” to the sudo group, greatening the privileges the user already had and more. This is vital to keep control over the GRUB menu and system security in general.

Task 3) Creating a Regular User Account for GRUB

```
matthew_tran@vm:~$ sudo adduser use
Adding user `use' ...
Adding new group `use' (1002) ...
Adding new user `use' (1002) with group `use' ...
Creating home directory `/home/use' ...
Copying files from `/etc/skel' ...
New password:
```

sudo adduser user

This command adds the user called “user.” The user will have less privileges compared to the “super” user, but it shows the difference in levels of access and power in the GRUB bootloader.

```
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for use
Enter the new value, or press ENTER for the default
        Full Name []:
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] Y
```

PWD = secured

Once more this command just sets a password/ another layer of defense.

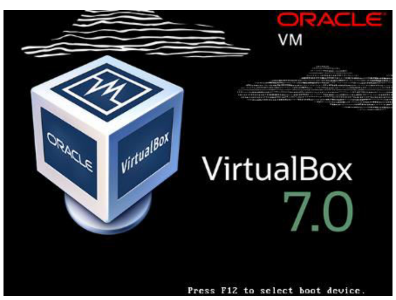
4) Verification and System Reboot

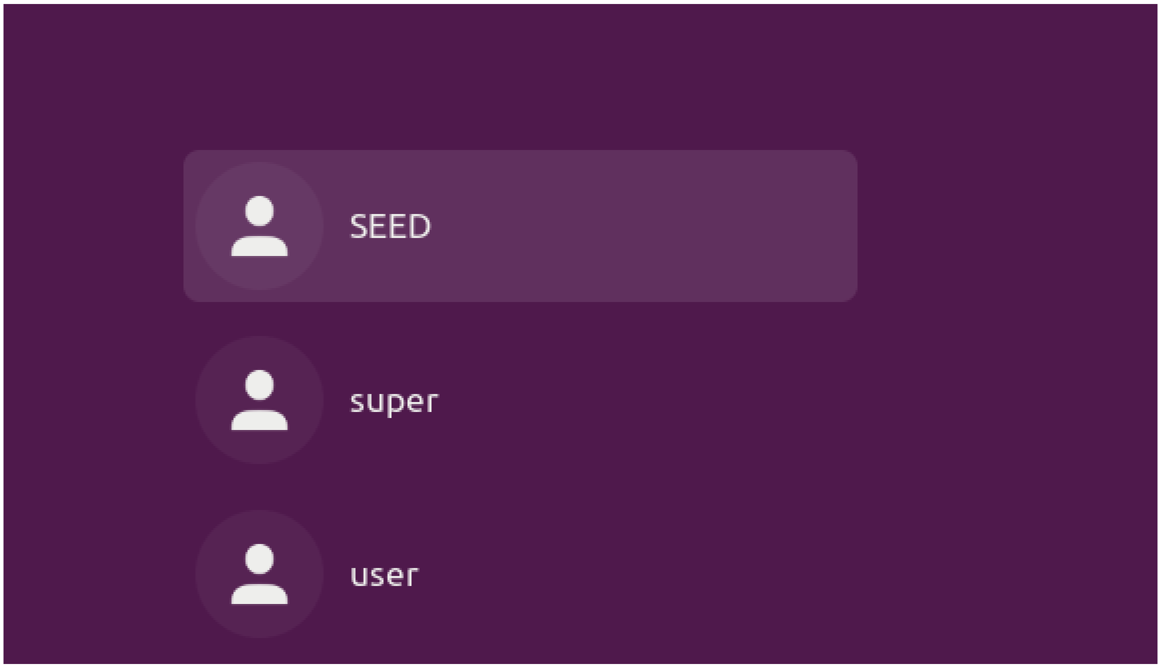
```
matthew_tran@vm:~$ groups super
super : super
```

groups super

This command will then check which group the user "super" belongs to. It will confirm whether or not the user is actually in the sudo group for administrative privileges or not.

```
matthew_tran@vm:~$ sudo reboot
```





sudo reboot

This reboots the system, allowing the changes made in the GRUB system to take place. GRUB system will only take effect after the system is rebooted.

```
[09/18/24]seed@VM:~$ sudo cat /etc/shadow | grep super
super:$6$FOPLnokvdtJL8gEP$WO4gx3uPbkzBM//qAT.STVshv5y2HnJY0l0pBsIHc
bSP1Ilu5NPG4mfK73VrWHQpP2nqgK/ss/lg53gYw2bsY/:19985:0:99999:7:::
```

sudo cat /etc/shadow | grep super

This command will check the /etc/shadow file for the user “super.” It will ensure the user was created. The password also starts with 6 which indicates the SHA-512 hashing algorithm, ensuring safety.

```
user:$6$o7djFJcSEAe3KyY1$ttYkcHSdqSiDMyH3Nqtk0sQamJ25YYaaegohwgll2s
yRGzh2Zl3L0waxU01ywfUI.ZOGIZRUKFWVlt7iaqYH01:19985:0:99999:7:::
```

sudo cat /etc/shadow | grep user

This has the same purpose, only this command was specified for the other user made, “user.”